WORKSHEET

Chapter 1: Equations of lines in space

1 Find the vector equation of the line through A and B:

a $A(2, -1, 3)$ and $B(1, 4, -1)$

b $A(1, 2, -2)$ and $B(2, -1, -1)$

c $A(1, 3, 2)$ and $B(2, -1, 4)$

d $A(2, -1, 3)$ and $B(1, 4, -3)$

2 Write down the vector equation of the line passing through the points with position vectors:

a $\mathbf{d} = 5\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{e} = 2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$

b $\mathbf{m} = 9\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ and $\mathbf{n} = 4\mathbf{i} + 5\mathbf{j} - \mathbf{k}$

c $\mathbf{p} = 3\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}$ and $\mathbf{q} = -\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$

d $\mathbf{r} = 2\mathbf{i} - \mathbf{j} + 5\mathbf{k}$ and $\mathbf{s} = 3\mathbf{i} + 6\mathbf{j} - 4\mathbf{k}$

3 Find the Cartesian equation of the line through A and B:

a $A(2, -1, 3)$ and $B(1, 4, -1)$

b $A(1, 2, -2)$ and $B(2, -1, -1)$

c $A(1, -3, 2)$ and $B(3, -1, 5)$

d $A(2, -1, 3)$ and $B(1, 4, -3)$

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e $A(9, 3, -2)$ and $B(4, 5, -1)$

f $A(2, -1, 5)$ and $B(3, 6, -4)$

4 Two lines have the vector equations:

$\mathbf{p} = -2\mathbf{i} + 3\mathbf{j} + \mathbf{k} + t(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$

$\mathbf{q} = 2\mathbf{j} + \mathbf{k} + s(-2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k})$

Determine if the lines are skew, parallel or intersect.

5 Show that the following two lines are parallel:

$\mathbf{p} = \mathbf{j} + 2\mathbf{k} + t(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$

$\mathbf{q} = -2\mathbf{i} - \mathbf{j} + s(-2\mathbf{i} - 4\mathbf{j} - 6\mathbf{k})$

6 **a** Find the vector equation of the line that passes through $(3, -1, 0)$ and is parallel to the line passing through $(-3, 2, 1)$ and $(2, 1, 0)$.

b Show that $P(-7, 1, 2)$ lies on the line determined in part **a**.

7 Consider the following vector equations:

Line 1: $\mathbf{p} = -2\mathbf{j} + 2\mathbf{k} + t(-\mathbf{i} + \mathbf{j})$

Line 2: $\mathbf{r} = -\mathbf{i} - \mathbf{j} + 2\mathbf{k} + s(-\mathbf{j} + \mathbf{k})$

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Show that Line 1 and Line 2 intersect and find their point of intersection.

8 Determine if the following lines are parallel or intersect. If the lines do intersect, find their point of intersection.

$\mathbf{d} = \mathbf{j} + 2\mathbf{k} + t(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$

$\mathbf{e} = -3\mathbf{i} + 4\mathbf{j} - 4\mathbf{k} + s(\mathbf{j} + 2\mathbf{k})$

9 Determine if the following lines are parallel, intersect or are skew. If the lines do intersect, find their point of intersection.

$\mathbf{m} = 2\mathbf{j} + \mathbf{k} + t(-2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k})$

$\mathbf{n} = -2\mathbf{i} + 3\mathbf{j} + \mathbf{k} + s(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$

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Answers

1 **a** $\mathbf{p} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k} + t(-\mathbf{i} + 5\mathbf{j} - 4\mathbf{k})$

b $\mathbf{p} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k} + t(\mathbf{i} - 3\mathbf{j} + \mathbf{k})$

c $\mathbf{p} = \mathbf{i} + 3\mathbf{j} + 2\mathbf{k} + t(\mathbf{i} - 4\mathbf{j} + 2\mathbf{k})$

d $\mathbf{p} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k} + t(\mathbf{i} - 5\mathbf{j} + 6\mathbf{k})$

2 **a** $\mathbf{p} = 5\mathbf{i} - 2\mathbf{j} + 3\mathbf{k} + t(-3\mathbf{i} + 3\mathbf{j} - 7\mathbf{k})$

b $\mathbf{p} = 9\mathbf{i} + 3\mathbf{j} - 2\mathbf{k} + t(-5\mathbf{i} + 2\mathbf{j} + \mathbf{k})$

c $\mathbf{p} = 3\mathbf{i} + 2\mathbf{j} + 7\mathbf{k} + t(-4\mathbf{i} - 4\mathbf{k})$

d $\mathbf{p} = 2\mathbf{i} - \mathbf{j} + 5\mathbf{k} + t(\mathbf{i} + 7\mathbf{j} - 9\mathbf{k})$

3 **a** $2 - x = \dfrac{y + 1}{5} = \dfrac{3 - z}{4}$

b $x - 1 = \dfrac{2 - y}{3} = z + 2$

c $\dfrac{x - 1}{2} = \dfrac{y + 3}{2} = \dfrac{z - 2}{3}$

d $x - 2 = \dfrac{y + 1}{3} = \dfrac{z - 3}{6}$

e $\dfrac{x - 9}{-5} = \dfrac{y - 3}{2} = z + 2$

f $x - 2 = \dfrac{y + 1}{7} = \dfrac{z - 5}{-9}$

4 The lines are parallel.

5 The lines are parallel.

6 **a** $3\mathbf{i} - \mathbf{j} + t(5\mathbf{i} - \mathbf{j} - \mathbf{k})$

b Equations are satisfied when $t = -2$.

7 The lines intersect at $(-1, -1, 2)$.

8 The lines intersect at $(-3, 4, -4)$.

9 The lines are skew.

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